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Master's Thesis

Sentinel

Automated Self-Belay System

An automated electronic climbing belay device that manages rope slack and performs fast-acting emergency braking, removing the dependency on a continuously attentive human belayer.

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Abstract

Sentinel is a mechatronic system designed to function as an automated electronic climbing belay device. Unlike conventional belay devices, which still require a human operator to feed rope and arrest falls, Sentinel aims to autonomously manage rope slack during a climber's ascent and to trigger a fast-acting mechanical brake upon detecting a fall. This thesis documents the design, implementation, and evaluation of a functional prototype, together with an analysis of the regulatory and safety context relevant to automated belay devices.

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Chapter 1

Introduction

The name *Sentinel* is derived from the Old Italian word *sentinella*, referring to a person or thing tasked with watching over, protecting, and alerting others of potential danger. This is precisely what the device aims to do for a climber: it continuously watches over the climbing rope, manages it to protect the climber during the ascent, and reacts immediately – by braking – the moment a danger (a fall) is detected.

1.1 Background

Rock and sport climbing rely fundamentally on the belay system: the mechanism and human partner responsible for managing rope slack and arresting a climber's fall. Traditional belaying requires a trained, attentive human belayer holding the rope at all times, which introduces risks related to human error, fatigue, distraction, and inconsistent reaction times. Semi-automated mechanical devices assist with braking by using a cam or pinching mechanism that clamps the rope against a grooved wheel, but they still require a human operator to feed rope in and out and to manage slack during the climb.

As an initial mechanical design reference, this project looks specifically at the Petzl GriGri, shown in Figure 1.1. Unlike a plain friction hitch or a static cam, the GriGri already relies on a mechanism with moving parts – a camming/pinching lever that clamps the rope against a grooved wheel to generate braking friction – which makes it a plausible starting point for turning into a mechatronic (i.e. motor-actuated and sensor-driven) system, rather than having to design a rope-gripping mechanism from scratch. A free 3D CAD model of this mechanism, available on the GrabCAD model repository [1], was used to study its mechanical layout and is used here purely as a starting point for the mechanical design of Sentinel's own rope-gripping mechanism, to be extended with active sensing and actuation as described in this thesis.

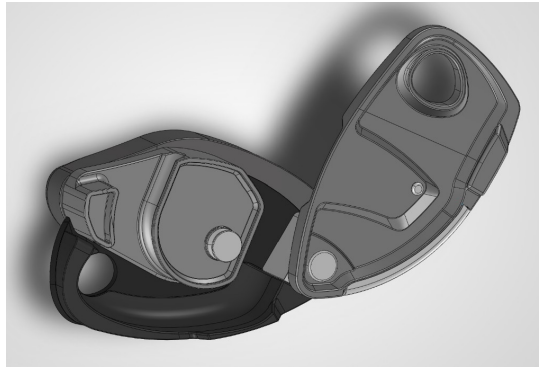


Figure 1.1: Petzl GriGri mechanical assembly. CAD model obtained from the GrabCAD model repository [1], used as a starting point for the design of Sentinel's rope-gripping mechanism.

1.2 Project Area

This thesis sits at the intersection of mechatronics, embedded real-time systems, and sports safety equipment engineering. The project area covers: the mechanical design of a rope-handling and braking mechanism; the integration of motion, position, and force sensing to estimate a climber's state in real time; the design of an active motor-driven mechanism for slack management and emergency braking; and the analysis of relevant safety regulations and standards for climbing protection equipment.

1.3 Narrowing

Given the breadth of the project area, this work narrows its focus to a single-rope, indoor, controlled-environment prototype. The system is narrowed to:

- A single, standard dynamic climbing rope diameter and type, rather than supporting the full range of ropes on the market.
- Bench-top and controlled indoor testing, rather than outdoor or production climbing-gym deployment.
- A functional proof-of-concept prototype, rather than a manufacturable, certified consumer product.

1.4 Principal Objectives

1. Create a functional prototype of an automated self-belaying device.
2. Perform comprehensive testing and evaluation of the prototype.
3. Analyze regulations, standards, and safety requirements applicable to such climbing devices.

1.5 Secondary Objectives

1. Survey existing commercial and academic self-belay and assisted-braking solutions.
2. Formulate comprehensive functional and technical requirement specifications for the system.

1.6 Research Questions

- RQ1** To what extent can an embedded sensing system reliably distinguish between a climber pulling in rope during normal ascent and the onset of an actual fall?
- RQ2** How can an automated belay mechanism safely transition out of a braked state once the climber has resumed a valid climbing position, without requiring manual intervention?
- RQ3** Which existing regulations and standards applicable to climbing protection equipment are relevant to an automated belay device, and what safety requirements do they impose on its design?
- RQ4** What mechanical and control-system safeguards are necessary to guarantee a safe fallback behaviour (e.g. automatic rope locking) in the event of a power or electronic failure?

1.7 Limitations

The following are explicitly outside the scope of this thesis, unless revisited as future work:

- Formal certification of the device against climbing safety standards (e.g. UIAA, CE/EN).
- Outdoor field testing with real climbers at height.
- Support for multiple rope diameters, types, or conditions (e.g. wet or icy rope).
- Weatherproofing and long-term durability testing of the prototype enclosure.
- Mass-manufacturability and industrial design for a commercial product.

1.8 Contributions

This thesis is expected to contribute:

1. A working proof-of-concept prototype demonstrating that automated, sensor-driven belaying with real-time slack management and fast emergency braking is technically feasible using accessible embedded hardware.

2. An analysis of existing commercial and academic self-belay solutions, situating this project's approach relative to the state of the art.
3. A documented functional and technical requirement specification for automated belay devices.
4. An analysis of applicable regulations, standards, and safety requirements relevant to automated belay device design.
5. Empirical performance data (e.g. reaction time, braking distance, false-trigger rate) characterising the prototype's safety-relevant behaviour under controlled test conditions.

1.9 Report Structure

This report is organised to build from problem definition to implementation and evaluation, following the structure of a typical software/mechatronic development-project thesis. Chapter 1 (this chapter) introduces the motivation, objectives, research questions, and scope. Chapter 2 defines the functional and non-functional requirements that the system must satisfy, at both a high and low level. Chapter 3 describes the overall architecture of the chosen solution and why it was chosen. Chapter 4 describes the development process followed, including tools for project management and documentation. Chapter 5 covers the technical details of the implementation (tools, languages, libraries, and representative code or design examples). Chapter 6 describes how the system is set up and deployed on the target hardware. Chapter 7 describes how the system was tested and what was learned from testing. Chapter 8 discusses the process and product choices made throughout the project, and Chapter 9 summarises what was achieved and outlines future work. Every chapter beyond this Introduction and the Requirements chapter is currently an empty placeholder, to be filled in as the project progresses; each structural change to this document is logged in the project's development log (`docs/DEV_LOG.md`).

1.10 Report Tools

This section documents the tools used in preparing this report and the underlying project work.

1.10.1 AI Tools

An AI-assisted coding and documentation environment (Cursor) was used throughout the project to accelerate firmware scaffolding, documentation drafting, and LaTeX authoring. All AI-assisted contributions were reviewed and directed by the author; ambiguous design decisions were resolved by the author rather than left to automated assumptions, per the project's development workflow.

1.10.2 CAD Tools

Mechanical design and reference modelling work uses SolidWorks, alongside publicly available reference CAD models (see Section 1.1) used for early-stage design inspiration.

1.10.3 LaTeX and Documentation Tools

This report is written in $\LaTeX$, using the community `ntnuthesis` document class for the long-form thesis, compiled locally with a MiKTeX distribution (`pdflatex/biber`). General project documentation (hardware specifications, pinouts, and development history) is maintained in Markdown under `docs/`.

1.10.4 Software Development Environments

Embedded firmware is developed in C++ using the PlatformIO build system and toolchain, targeting an embedded microcontroller platform. Version control is managed with Git.

Chapter 2

Functional Requirements

This chapter defines the functional requirements that the Sentinel system must satisfy. Requirements are described functionally, independent of the specific hardware components used to realise them; component selection and justification are deferred to dedicated design chapters later in this report.

2.1 Climbing Mechanics and Slack Management

The system must automatically take in rope as the climber ascends, so that the amount of slack in the system remains within a safe, comfortable range without manual intervention. To do so, the system must reliably distinguish between two dynamically different situations that both involve rope moving through the mechanism:

- **Normal ascent / rope pull-in**, where the climber is moving upward and the rope must be taken in at a corresponding, comparatively gradual rate.
- **A fall event**, characterised by a sudden increase in rope acceleration and/or a rapid increase in measured rope tension, which must be distinguished from normal climbing dynamics (e.g. clipping, resting on the rope, or bursts of quick movement).

This distinction must be made using onboard rope-movement and tension-sensing information, processed in real time, and must be robust enough to avoid both missed fall detections and false-positive brake triggers during ordinary climbing.

2.2 Post-Fall State Detection

After the system has arrested a fall by engaging the brake, it must be able to detect when the climber has resumed a valid, stable climbing position (e.g. has regained a hold and is no longer hanging dynamically on the rope). Once this state is detected, the system must release the brake and resume normal slack-management

behaviour automatically, without requiring a manual reset from the climber or a belay partner, unless a manual reset is explicitly designed as a deliberate safety fallback.

2.3 Operational Modes and Remote Control

The system must support at least two distinct operational modes:

- **Climbing Mode:** the default mode described in Section 2.1 and Section 2.2, in which the system manages slack during ascent and arrests falls.
- **Rappel / Abseil Mode:** a mode that allows a controlled, continuous descent, with the braking behaviour adapted for a deliberate downward motion rather than treating it as a fall event.

Switching between these modes must be possible via a wireless remote control, allowing the operator (climber or belay supervisor) to change modes without physically accessing the device mounted on the anchor point.

2.4 Battery Management and Safety Protocols

2.4.1 Battery Level Indication

The system must visually and clearly indicate the current battery charge level at all times during operation, so that the user can make an informed decision about whether it is safe to begin or continue climbing.

2.4.2 Low-Battery Restrictions

The system must enforce the following safety protocols based on battery state:

- If the battery level is low at start-up, the system must prevent Climbing Mode from being initiated.
- If the battery level drops to a low or critical level *during* an active climbing session, the system must restrict its functionality strictly to Rappel/Abseil Mode, allowing the climber currently on the wall to be safely lowered rather than left in an unsupported state.

2.5 Fail-Safe Mechanical Override

This requirement is considered **critical** to the safety case of the system.

2.5.1 Automatic Rope Lock on Power Loss

In the event of a power loss or an electronic/software failure, the rope must lock automatically by default. The mechanical design must guarantee this fail-safe be-

haviour independent of the control software – that is, the default, unpowered, or faulted state of the mechanism must be “locked”, not “free-running”.

2.5.2 Manual Override Lever

A manual physical lever must allow the climber (or a belay supervisor) to override the automatic lock and execute a controlled descent by hand, even with no power available. This requires evaluating a passive, GriGri-like mechanical behaviour that functions correctly without any electronic assistance, ensuring that a power or electronics failure never leaves a climber stranded with no means of controlled descent.

2.6 Non-Functional Requirements

Beyond the functional behaviour described above, the system must also satisfy a number of non-functional requirements. These are not yet exhaustive and are expected to be expanded as the design matures; initial candidates include:

- **Reliability and safety-critical performance:** given that a missed or delayed fall detection has direct safety consequences, the system’s sensing and braking response must meet a bounded, worst-case reaction latency, and must degrade towards the fail-safe locked state (Section 2.5) rather than towards an unsafe state under any fault condition.
- **Real-time responsiveness:** sensor acquisition, state estimation, and brake actuation must run within firm real-time deadlines, since the usefulness of the system depends on reacting fast enough to arrest a fall before excessive slack develops.
- **Usability and ergonomics:** the device must be operable by a climber with no more training than is required for a conventional assisted-braking belay device, with clear, unambiguous feedback (e.g. battery level, operating mode) that does not require reading a manual mid-climb.

Chapter 3

Technical Design

3.1 Conceptual Design Sketch (Phase 1)

Figure 3.1 shows an early-stage hand sketch produced during Phase 1 of the mechanical design, used to visualise which functions the rope-pinching mechanism needs to provide before committing to a detailed CAD model. The sketch identifies the following elements:

- An **encoder**, used to measure the rope's acceleration as it moves through the mechanism.
- A **load pin**, integrated into the pivot axle of the pinching lever, used to measure the tension on the climbing side of the rope (i.e. the side leading up to the climber). This tension reading provides the information needed to determine when the climber has regained a stable position on the wall, so that the pinch can be safely released (see Section 2.2).
- A **spring** built into the pinching mechanism, providing the passive fail-safe behaviour described in Section 2.5: the mechanism defaults to a locked/pinched state on its own, without requiring power.
- A **servo**, used to actuate the pinching lever – i.e. to open and close the brake under motor control.
- **Axles for the outer cover**, and a **hole for the carabiner** used to attach the device to the anchor point.
- An **accommodation volume for the electronics**, i.e. the internal space reserved for the control board and wiring.
- A **potentiometer**, mounted in the pinching mechanism, used to sense the lever's current angular position (i.e. how open or closed the brake currently is).

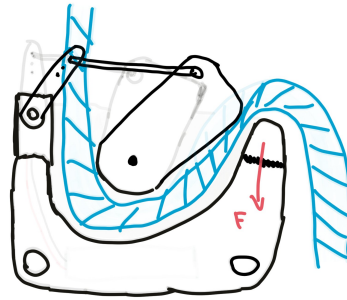


Figure 3.1: Phase 1 conceptual sketch of the rope-pinching mechanism, used to visualise the device's required functions.

Chapter 4

Development Process

Chapter 5

Implementation

Chapter 6

Deployment

Chapter 7

Testing and User Feedback

Chapter 8

Discussion

Chapter 9

Conclusion

Bibliography

- [1] GrabCAD Community, *Petzl GriGri dummy – CAD model*, GrabCAD, 3D CAD reference model used as the mechanical design inspiration for the Sentinel rope-gripping mechanism. Author/upload date not yet verified – update after checking the source page. Accessed: 25th Aug. 2026. [Online]. Available: https://grabcad.com/library/petzl_grigri_dummy-1